MONTULAR
Montelukast Sodium Tablets 10 mg

GENERAL CHARACTERISTIC

General physico-chemical properties

Beige colored, round shape, biconvex, film coated tablets, plain on both sides.

COMPOSITION

Each film coated tablet contains
Montelukast Sodium equivalent to Montelukast USP…………10 mg

Excipients: Microcrystalline cellulose (Avicel PH102), croscarmellose sodium, sodium lauryl sulphate, magnesium stearate, purified water and opadry 20A520035 yellow.

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Leukotriene receptor antagonists
ATC Code: R03D C03

Mechanism of action

The cysteinyl leukotrienes (LTC_4, LTD_4, LTE_4) are potent inflammatory eicosanoids released from various cells including mast cells and eosinophils. These important pro-asthmatic mediators bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 ($CysLT_1$) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other pro-inflammatory cells (including eosinophils and certain myeloid stem cells). CysLTs have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include bronchoconstriction, mucous secretion, vascular permeability, and eosinophil recruitment. In allergic rhinitis, CysLTs are released from the nasal mucosa after allergen exposure during both early- and late-phase reactions and are associated with symptoms of allergic rhinitis. Intranasal challenge with CysLTs has been shown to increase nasal airway resistance and symptoms of nasal obstruction.

Pharmacodynamic effects

Montelukast is an orally active compound which binds with high affinity and selectivity to the $CysLT_1$ receptor. In clinical studies, montelukast inhibits bronchoconstriction due to inhaled LTD_4 at doses as low as 5 mg. Bronchodilation was observed within 2 hours of oral administration. The bronchodilation effect caused by a beta-agonist was additive to that caused by montelukast. Treatment with montelukast inhibited both early- and late-phase bronchoconstriction due to antigen challenge. Montelukast, compared with placebo, decreased peripheral blood eosinophils in adult and paediatric patients. In a separate study, treatment with montelukast significantly decreased eosinophils in the airways (as measured in sputum) and in peripheral blood while improving clinical asthma control.

PHARMACOKINETIC PROPERTIES

Absorption

Montelukast is rapidly absorbed following oral administration. For the 10-mg film-coated tablet, the mean peak plasma concentration (C_{max}) is achieved 3 hours (T_{max}) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and C_{max} are not influenced by a standard meal. Safety and efficacy were demonstrated in clinical trials where the 10mg film-coated tablet was administered without regard to the timing of food ingestion.

Distribution

Montelukast is more than 99% bound to plasma proteins. The steady state volume of distribution of montelukast averages 8-11 liters. Studies in rats with radiolabelled montelukast indicate minimal distribution across the blood-brain barrier. In addition, concentrations of radiolabelled material at 24 hours postdose were minimal in all other tissues.

Biotransformation

Montelukast is extensively metabolized. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children.

Cytochrome P450 2C8 is the major enzyme in the metabolism of montelukast. Additionally, CYP 3A4 and 2C9 may have a minor contribution, although itraconazole, an inhibitor of CYP 3A4, was shown not to change pharmacokinetic variables of montelukast in healthy subjects that received 10 mg montelukast daily.

In vitro studies using human liver microsomes indicate that cytochrome P450, 3A4, 2A6 and 2C9 are involved in the metabolism of montelukast. Based on further *in vitro* results in human liver microsomes, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450, 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6. The contribution of metabolites to the therapeutic effect of montelukast is minimal.

Elimination

The plasma clearance of montelukast averages 45 ml/min in healthy adults. Following an oral dose of radiolabeled montelukast, 86% of the radioactivity was recovered in 5 day fecal collections and <0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively via the bile.

Characteristics in Patients

No dosage adjustment is necessary for the elderly or mild to moderate hepatic insufficiency. Studies in patients with renal impairment have not been undertaken. Because montelukast and its metabolites are eliminated by the biliary route, no dose adjustment is anticipated to be necessary in patients with renal impairment. There are no data on the pharmacokinetics of montelukast in patients with severe hepatic insufficiency (Child-Pugh score >9).

With high doses of montelukast (20 and 60 fold the recommended adult dose), decrease in plasma theophylline concentration was observed. This effect was not seen at the recommended dose of 10 mg once daily.

INDICATIONS

For the prophylaxis and chronic treatment of asthma in adults and adolescents 15 years of age and older. MONTULAR is indicated in adults and adolescents 15 years of age and older for the relief of daytime and night-time symptoms of seasonal allergic rhinitis.

DOSAGE AND MODE OF ADMINISTRATION

MONTULAR should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs.

Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

Adults 15 Years of Age and Older with Asthma and/or Seasonal Allergic Rhinitis.

The dosage for adults 15 years of age and older is one 10-mg tablet daily.

General Recommendations

The therapeutic effect of MONTULAR on parameters of asthma control occurs within one day. MONTULAR tablets can be taken with or without food. Patients should be advised to continue taking MONTULAR while their asthma is controlled, as well as during periods of worsening asthma.

No dosage adjustment is necessary for the elderly, for patients with renal insufficiency, or mild-to-moderate hepatic impairment, or for patients of either gender.

Montelukast is a long term-controller medication which should not be substituted for short acting beta-agonists. It is effective alone or in combination with other prophylactic agent.

Montelukast is a preventive agent, which should be used in addition to other drugs for the management of asthma.

Therapy with MONTULAR in Relation to Other Treatments for Asthma MONTULAR can be added to a patient's existing treatment regimen.

Reduction in Concomitant Therapy:

Bronchodilator Treatments: MONTULAR can be added to the treatment regimen of patients who are not adequately controlled on bronchodilator alone. When a clinical response is evident (usually after the first dose), the patient's bronchodilator therapy can be reduced as tolerated.

Inhaled Corticosteroids: Treatment with MONTULAR provides additional clinical benefit to patients treated with inhaled corticosteroids. A reduction in the corticosteroid dose can be made as tolerated. The dose should be reduced gradually with medical supervision. In some patients, the dose of inhaled corticosteroids can be tapered off completely. MONTULAR should not be abruptly substituted for inhaled corticosteroids.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients listed above.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled β-agonist should be used. Patients should seek their doctors' advice as soon as possible if they need more inhalations of short-acting β-agonists than usual.

Montelukast should not be substituted abruptly for inhaled or oral corticosteroids.

There are no data demonstrating that oral corticosteroids can be reduced when montelukast is given concomitantly.

In rare cases, patients on therapy with anti-asthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases have been sometimes associated with the reduction or withdrawal of oral corticosteroid therapy. Although a causal relationship with leukotriene receptor antagonism has not been established, physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.

Treatment with montelukast does not alter the need for patients with aspirin-sensitive asthma to avoid taking aspirin and other non-steroidal anti-inflammatory drugs.

INTERACTIONS WITH OTHER MEDICINAL PRODUCTS

Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following medicinal products: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl estradiol/ norethindrone 35/1), terfenadine, digoxin and warfarin.

The area under the plasma concentration curve (AUC) for montelukast was decreased approximately 40% in subjects with co-administration of phenobarbital. Since montelukast is metabolised by CYP 3A4, 2C8, and 2C9, caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, 2C8, and 2C9, such as phenytoin, phenobarbital and rifampicin.

In vitro studies have shown that montelukast is a potent inhibitor of CYP 2C8. However, data from a clinical drug-drug interaction study involving montelukast and rosiglitazone (a probe substrate representative of medicinal products primarily metabolized by CYP 2C8) demonstrated that montelukast does not inhibit CYP 2C8 in vivo. Therefore, montelukast is not anticipated to markedly alter the metabolism of medicinal products metabolised by this enzyme (e.g., paclitaxel, rosiglitazone, and repaglinide).

In vitro studies have shown that montelukast is a substrate of CYP 2C8, and to a less significant extent, of 2C9, and 3A4. In a clinical drug-drug interaction study involving montelukast and gemfibrozil (an inhibitor of both CYP 2C8 and 2C9) gemfibrozil increased the systemic exposure of

montelukast by 4.4-fold. No routine dosage adjustment of montelukast is required upon co-administration with gemfibrozil or other potent inhibitors of CYP 2C8, but the physician should be aware of the potential for an increase in adverse reactions.
Based on in vitro data, clinically important drug interactions with less potent inhibitors of CYP 2C8 (e.g., trimethoprim) are not anticipated. Co-administration of montelukast with itraconazole, a strong inhibitor of CYP 3A4, resulted in no significant increase in the systemic exposure of montelukast.

PREGNANCY AND LACTATION

Pregnancy**:** Animal studies do not indicate harmful effects with respect to effects on pregnancy or embryonal/fetal development.
Limited data from available pregnancy databases do not suggest a causal relationship between montelukast and malformations (i.e. limb defects) that have been rarely reported in worldwide post marketing experience.
Montular may be used during pregnancy only if it is considered to be clearly essential.

Lactation: Studies in rats have shown that montelukast is excreted in milk. It is not known if montelukast is excreted in human milk.
Montular may be used in nursing mothers only if it is considered to be clearly essential.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Montular is not expected to affect a patient's ability to drive a car or operate machinery. However, in very rare cases, individuals have reported drowsiness or dizziness.

UNDESIRABLE EFFECTS

The following drug-related adverse reactions in clinical studies were reported commonly (≥1/100 to <1/10) in asthmatic patients treated with montelukast and at a greater incidence than in patients treated with placebo:

Body System Class	**Adult Patients 15 years and older (two 12-week studies; n=795)**	**Paediatric Patients 6 to 14 years old (one 8-week study; n=201) (two 56-week studies; n=615)**
Nervous system disorders	headache	headache
Gastrointestinal disorders	abdominal pain	

With prolonged treatment in clinical trials with a limited number of patients for up to 2 years for adults, and up to 12 months for paediatric patients 6 to 14 years of age, the safety profile did not change.

Post-marketing Experience

Adverse reactions reported in post-marketing use are listed by System Organ Class and specific adverse experience term below. Frequency categories were estimated based on relevant clinical trials.

Infections and infestations

Very common : Upper respiratory infection[@]

Blood and lymphatic system disorders

Rare : increased bleeding tendency

Immune system disorders

Uncommon : Hypersensitivity reactions including anaphylaxis

Very Rare : hepatic eosinophilic infiltration.

Psychiatric disorders

Uncommon : Dream abnormalities including nightmares, insomnia, somnambulism, irritability, anxiety, restlessness, agitation including aggressive behaviour or hostility, depression, psychomotor hyperactivitiy (including irritability, restlessness, tremor[§])

Rare : Disturbance in attention, memory impairment

Very rare : Hallucinations, disorientation, suicidal thinking and behaviour (suicidality)

Obsessive Compulsive Symptoms

Nervous system disorders

Uncommon : Dizziness, drowsiness paraesthesia/hypoesthesia, seizure

Cardiac disorders

Rare : Palpitations

Respiratory system, thoracic and mediastinal disorders

Uncommon : Epistaxis

Very Rare : Churg-Strauss Syndrome (CSS), Pulmonary eosinophilia

Gastrointestinal disorders

Uncommon : Diarrhoea**, nausea**, vomiting**

Common : Dry mouth, dyspepsia

Hepato-biliary disorders

Common : Elevated levels of serum transaminases (ALT, AST)

Very Rare : Hepatitis (including cholestatic, hepatocellular, and mixed-pattern liver injury).

Skin and subcutaneous tissue disorders

Common : Rash**

Uncommon : Bruising, urticaria, pruritus

Rare : Angiooedema

Very Rare : Erythema nodosum erythema multiforme

Musculoskeletal and connective tissue disorders

Uncommon : Arthralgia, myalgia including muscle cramps

General disorders and administration site conditions

Common : Pyrexia**

Uncommon : Asthenia/fatigue, malaise, oedema

* Frequency Category: Defined for each Adverse Experience Term by the incidence reported in the clinical trials data base: Very Common (≥1/10), Common (≥1/100 to <1/10), Uncommon (≥1/1000 to <1/100), Rare (≥1/10,000 to <1/1000), Very Rare (<1/10,000).

@ This adverse experience, reported as very common in the patients who received montelukast, was also reported as very common in the patients who received placebo in clinical trials.

**This adverse experience, reported as common in the patients who received montelukast, was also reported as common in the patients who received placebo in clinical trials.

§ Frequency Category: Rare

Postmarketing experience

Blood and lymphatic system disorders: thrombocytopenia

OVERDOSE AND TREATMENT

In chronic asthma studies, montelukast has been administered at doses up to 200 mg/day to adult patients for 22 weeks and in short term studies, up to 900 mg/day to patients for approximately one week without clinically important adverse experiences.

There have been reports of acute overdose in post-marketing experience and clinical studies with montelukast. These include reports in adults and children with a dose as high as 1,000 mg (approximately 61 mg/kg in a 42-month old child). The clinical and laboratory findings observed were consistent with the safety profile in adults and paediatric patients. There were no adverse experiences in the majority of overdose reports.

Symptoms of overdose

The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.

Management of overdose

No specific information is available on the treatment of overdose with montelukast. It is not known whether montelukast is dialysable by peritoneal- or haemodialysis.

STORAGE CONDITION

Store in dry place, temperature below 30°C. Keep all medicines out of reach of children.

DOSAGE FORM AND PACKAGING AVAILABLE
Alu- alu blister pack 10's (Box of 30's).

NAME AND ADDRESS OF MANUFACTURER
Kusum Healthcare Pvt. Ltd.
SP-289(A), RIICO Industrial Area,
Chopanki, Bhiwadi, Alwar,
(Rajasthan), India

PRODUCT REGISTRATION HOLDER
Synerrv Sdn Bhd,
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DATE OF REVISION OF PACKAGE INSERT
May 2019